

Jeric Lew Jieyi

Research Assistant (Robotics)

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Education

National University of Singapore (NUS), Singapore

Master of Engineering (Mechanical Engineering) | GPA: 5.0/5.0 Aug 2025 – May 2027

Bachelor of Engineering (Mechanical Engineering) | GPA: 4.9/5.0 Aug 2021 – May 2025

Robotics Specialisation; Minor in Computer Science; Minor in Innovation and Design Program

Georgia Institute of Technology, Atlanta, GA

Undergraduate Study Abroad (Non-Degree) | GPA: 4.0/4.0 Jan 2024 – May 2024

Relevant Coursework: *Emerging Topics in AI and Deep Learning; Deep Learning for Robotics; Robotics; Robotics System Design; Perception & Robotics; Soft Robotics; Microprocessor Applications*

Research & Professional Experience

MARMot Lab, National University of Singapore (NUS), Singapore

Research Assistant & Master's Thesis with [Guillaume Sartoretti](#) Aug 2025 – Present

- Research on unsupervised RL (skill discovery) with World Models for single/multi-agent systems
- Mentoring undergraduate students projects on diffusion models and RL for path planning

Research Intern & Bachelor's Thesis with [Guillaume Sartoretti](#) & [Cao Yuhong](#) Aug 2023 – Jul 2025

- Research on diffusion models and RL for path planning in single/multi-agent robot systems
- Developed and tested novel deep-learning methods in high fidelity simulations and real robots

AirLab, Carnegie Mellon University (CMU), Pittsburgh, PA

Robotics Research Intern with [Sebastian Scherer](#) & [Wenshan Wang](#) Jun 2024 – Aug 2024

- Research on [off-road navigation](#) as a part of CMU's [Robotics Institute Summer Scholars](#) program
- Distilled vision foundation models (DINOv2, RADIO) for faster inference and higher feature resolution
- Integrated pipeline into a large, complex, multi-language off-road navigation stack with ROS

DSO National Laboratories, Singapore

Robotics Intern May 2023 – Aug 2023

- Deployed real-time object detection (YOLOv5) and tracking (DeepSORT) on Jetson Xavier NX
- Integrated cameras into a robotic system by developing ROS2 software drivers

Selected Publications

AID: Agent Intent from Diffusion for Multi-Agent Informative Path Planning

Jeric Lew, Yuhong Cao, Derek Ming Siang Tan, Guillaume Sartoretti

International Conference on Swarm Intelligence (ANTS), 2026 | [Code](#)

DARE: Diffusion Policy for Autonomous Robot Exploration

Yuhong Cao*, Jeric Lew*, Jingsong Liang, Jin Cheng, Guillaume Sartoretti

IEEE International Conference on Robotics and Automation (ICRA), 2025 | [Code](#) | [Video](#)

SALON: Self-supervised Adaptive Learning for Off-road Navigation

Matthew Sivaprakasam, Samuel Triest, Cherie Ho, Shubhra Aich, Jeric Lew, Isaiah Adu, Wenshan Wang, Sebastian Scherer

IEEE International Conference on Robotics and Automation (ICRA), 2025 | [Project](#) | [Video](#)

SHRED: Swift High-Resolution features via Efficient Distillation

Jeric Lew, Matthew Sivaprakasam, Samuel Triest, Wenshan Wang, Sebastian Scherer

RISS Working Papers Journal, 2024 | [Poster](#) | [Video](#)

A novel application for real-time arrhythmia detection using YOLOv8

Guang Jun Nicholas Ang, Aritejh Kr Goil, Henryk Chan, **Jieyi Jeric Lew**, Xin Chun Lee,

Raihan Bin Ahmad Mustaffa, Timotius Jason, Ze Ting Woon, Bingquan Shen

arXiv, 2024

Skills

Programming Languages: Python, C/C++, MATLAB, Java, ARM-7 Assembly

Frameworks/Tools: PyTorch, ROS1/2, MuJoCo, CUDA, OpenCV, Arduino

Hardware: SOLIDWORKS, 3D Printing, Machining, Microcontroller, Blender

Concepts: Reinforcement Learning (PPO, SAC, etc.), Imitation Learning (Diffusion, Flow Matching), Computer Vision (UNet, ViT, DINOv2), Planning (A*, D*, RRT*, etc.)

Selected Projects

PPO for Block Pushing Task with Robot Arm

github.com/JericLew/Push_MuJoCo

Graduate robotics class to leverage DRL for any robotics task

- Designed a custom MuJoCo simulation environment for specific block pushing task
- Designed neural network architecture for privileged state inputs and more realistic vision inputs
- Leveraged PPO with behaviour cloning to train a vision-based policy from pre-trained privileged policy

Robotics System Design ft. TurtleBot

github.com/Magmanat/r2auto_nav

Undergraduate robotics class with the task to traverse and map a maze and then locating a hot target and firing projectiles

- Sized and chose actuators and sensors for chose design based on literature review and calculations.
- Interfaced thermal camera and NFC reader with ROS2
- Developed navigation and target seeking algorithm using Python with ROS2 and Linux environment

Deep-Learning Pose Estimation for Sports Training

github.com/JasonYapzx/sportform

Hackathon entry aimed to promote healthy living by gamifying exercises

- Utilised deep-learning computer vision algorithm (YOLOv8) for human pose estimation to locate joints
- Developed Python scripts with OpenCV to count exercise repetitions and check form of exercise

Teaching

Teaching Assistant, [ME1102 Engineering Principles and Practice I](#)

Fall 2024

Teaching Assistant, [EG1311 Design and Make](#)

Summer 2022 - Fall 2023

Teaching Assistant, [CS1010E Programming Methodology](#)

Fall 2022

Awards and Honors

[Best Student in Microprocessor Applications](#)

Spring 2023

Dean's List

Fall **2022**, **2023**, **2024**

NUS Merit Scholarship

Fall 2021 - Spring 2025